

Grid Optimized Operation Dispatch (GOOD) Model Formulation

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1 Introduction

The Grid Optimized Operation Dispatch (GOOD) model is a high level power grid constrained Economic Dispatch (OD) and Capacity Expansion (CE). The purpose of the GOOD model is to provide a flexible open-source model for the evaluation of energy technology and policy. ...

1.1 Structure

The first fundamental unit of the GOOD model is the balancing region. A balancing region contains assets (generators, loads, and storage). Assets may only belong to a single region. Balancing regions are connected to one-another via transmission edges containing a set of lines. Regional net energy must be zero at each time step (this may be a hard or soft constraint). The second fundamental unit of the GOOD model is the policy. Policies apply to groups of assets which may include assets from multiple balancing regions. Multiple policies may apply to a single asset. Assets may belong to multiple policies. The GOOD model contains two fundamental components:

1. **Energy System Graph** The energy system is defined by the directed graph $G = \{R, E\}$ where R is the set of regions and E is the set of transmission connections between regions. Each node $r \in R$ has a set of assets $A_r \subseteq A$ where A is the set of assets in the model. Each edge $(i, j) \in E$ has a set of lines $L_{(i,j)} \subseteq L$ where L is the set of lines in the model.
2. **Policy Set** Policies are contained in the set P where each policy $p \in P$ applies to a set of assets $A_p \subseteq A$.

The objective of the GOOD model is to produce an optimal set of operational (OD) and strategic (CE) variables such that overall cost is minimized subject to constraints. Constraints are at the asset level (maximum output, ramp rate, maximum capacity, etc.), the energy system level (energy balance, transmission limits), and the governance level (renewable profile).

1.2 Formulation

1.2.1 Objective Function

$$\min_{X \in \bar{X}, Z \in \bar{Z}} J(X, Z) = J_X(X) + J_Z(Z) \quad (1)$$

$$J_X(X) = \sum_{t \in T} \left[\sum_{r \in R} [c_r^s x_{r,t}^s + c_r^w x_{r,t}^w] + \sum_{p \in P} c_p^c x_{p,t}^c + \sum_{a \in A} \sum_{x_i^a \in X_a} c_i^a x_{i,t}^a + \sum_{l \in L} \sum_{x_i^l \in X_l} c_i^l x_{i,t}^l \right] \quad (2)$$

$$J_Z(Z) = \sum_{d \in D} \left[\sum_{a \in A} \sum_{z_j^a \in Z_a} c_j^a z_{j,d}^a + \sum_{l \in A} \sum_{z_j^l \in Z_a} c_j^l z_{j,d}^l \right] \quad (3)$$

The overall cost function, expressed in (1), minimizes operational costs J_X as a function of operational variables X and strategic costs J_Z as a function of strategic variables Z . Operational costs, expressed in (2), are the sum across time-steps T of the sum of regional shortfall x_s and wastage x_w , policy compliance x_c , asset operations x_i^a and line operations x_i^l . Shortfall and wastage are optional variables which serve to penalize failure to balance energy in a region and, thus, allow for soft enforcement of the regional balancing constraint. Compliance costs serve to penalize failing to meet a policy constraint and allow for soft enforcement of the constraint. Operations for assets and lines include variables such as energy production, consumption, and transmission. Strategic costs, expressed in (3), are the sum across investment periods D of asset capacity expansion z_j^a and line capacity expansion z_j^l . Assets and lines may have multiple operational and strategic variables, a single operational and/or strategic variable, or none of either.

1.2.2 Regional Energy Balance

$$\sum_{a \in A_r} \sum_{x_i^a \in X_a} x_{i,t}^a + \sum_{\theta \in \Theta_r} \sum_{l \in L(\theta,r)} \eta^l x_t^l - \sum_{\phi \in \Phi_r} \sum_{l \in L(r,\phi)} x_t^l + x_t^s - x_t^w = 0 \quad \forall t \in T, r \in R \quad (4)$$

Θ_r is the set of nodes with transmission connections in to r , Φ_r is the set of nodes with transmission connections out of r

1.2.3 Renewable Profile Standard

$$\sum_{a \in A_{P,I}} y_d^a - \rho \sum_{a \in A_P} y_d^a \geq 0 \quad (5)$$

1.2.4 Domain of Variables

$$X, Z \in \mathbb{R} \quad (6)$$

1.3 Implementation